

**PROGRAMME: BCA (Bachelor of Computer Applications)
SEMESTER – I**

Teaching-Learning & Evaluation Plan

Course Information: Core Course

Course Code: 23BCA1C05	Course Title: Fundamentals of Computer Applications
Credits Units: 03	Contact Hours: 45 L-T-P: 3-0-0
CA: UE Weightage - 50:50	Pass Marks (CA & UE) – 20:20 Aggregate Pass Marks: 40
UE Question Paper Marks: 50	Special Examination Fees: NA Pre-requisite (if any): NA
Course Facilitator (s): Prof. Neetha S.S, Assistant Professor	

Programme Outcomes (POs)	
At the end of the programme, students will be able to	
PO 1	Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.
PO 2	Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.
PO 3	Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand, and propose integrated solutions using emerging technologies.
PO 4	Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well-informed conclusions.
PO 5	Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions.
PO 6	Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.
PO 7	Life-long Learning: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.
PO 8	Project Management: Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.

PO 9	Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.
PO 10	Societal & Environmental Concern: Ability to recognize economic, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.
PO1 1	Individual & Team Work: Ability to work as a member or leader in diverse teams in multidisciplinary environment.
PO1 2	Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.
Program Specific Outcomes (PSO's)	
PSO 1	Pertain current knowledge and adapting to emerging applications of Mathematics, Science fundamentals in the field of Computer science and its applications.
PSO 2	Exhibit proficiency in identifying, formulating and analyzing complex problems in the computer environment.
PSO 3	Ability to create, select and apply appropriate modern techniques for solving complex issues.
PSO 4	Explore technical knowledge in diverse areas of Computer Applications and experience a conducive environment in nurturing skills for successful career and higher studies.

Course Objectives (COB's)

COB1: Use MS Office software for word-processing, data analysis and preparing presentations.

COB2: Develop static web pages using HTML

Course Outcomes (CO's)

At the end of the course, students will be able to

CO1: Demonstrate the use of file management techniques for file and directory/folder organization.

CO2: Create a document using MS Word appropriate for the solution

CO3: Solve basic information systems problems by applying spreadsheet techniques.

CO4: Solve basic information systems problems by applying presentation software techniques.

CO5: Develop static web pages.

Sl. No.	Course Outcome	Skill's	Pos	Bloom's Taxonomy Level
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1.	CO 1	Understanding	PO1, PO2, PO3	L2
2.	CO 2	Apply, Analyze and Evaluate	PO1, PO2, PO3, PO4, PO7	L3, L4, L5
3.	CO 3	Apply, Analyze and Evaluate	PO1, PO2, PO3, PO4, PO7.	L3, L4, L5
4.	CO 4	Apply, Analyze	PO1, PO2, PO3, PO4, PO7, PO9	L3, L4
5.	CO 5	Understanding	PO1, PO2, PO3, PO4, PO5, PO7, PO12.	L2

CO-PO/PSO Mapping:

Course Outcome	Blooms Taxonomy Levels	Program Outcomes (PO's) Mapped												Program Specific Outcomes (PSO's)			
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
														L3	L4	L6	L4
CO1	L2	1	1	2	-	-	-	-	-	-	-	-	-	3	1	1	1
CO2	L5	3	3	3	3	-	-	3	-	-	-	-	-	3	3	3	1
CO3	L5	3	3	3	3	-	-	2	-	2	-	3	-	3	3	3	2
CO4	L4	2	2	2	2	2	-	-	2	2	-	2	-	2	2	2	2
CO5	L2	1	-	-	2	2	2	2	-	1	2	1	-	2	2	2	2

Course Contents:

Module	Details	Contact Hours
I	Basics of Computer System: Concept of Hardware and Software, Computer block diagram, Input Output unit, CPU, Control Unit, Arithmetic logic Unit (ALU), Memory Unit, Monitor. Printers: Dot matrix, Laser, Inkjet, Plotters, Scanner. System software and Application Software, Operating system concepts,	9

	purpose and functions, Operations of Windows OS. Creating and naming of file and folders, copying file, renaming and deleting of files and folders, searching files and folders, installation application, creating shortcut of application on the desktop Overview of control Panel, Taskbar.	
II	Using MS - Word (Latest Version): Overview of Word processor, Basics of Font type, size, color, Effects like Bold, italic, underline, Subscript and superscript, Case changing options, Inserting, deleting, undo and redo, Copy and Moving (cutting) text within a document, Formatting Paragraphs and Lists, Setting line spacing; single, Page settings and margins including header and footer, Spelling and Grammatical checks, Table and its options, Inserting rows or columns, merging and splitting cells, Arithmetic Calculations in a Table, Working with pictures, Inserting Pictures from Files, Using Drawings and WordArt; Lines and Shapes, Modifying Drawn Objects, Formatting Drawn Objects, options for Creating and Modifying a WordArt Object	9
III	Using MS - Excel (Latest Version): Introduction to Excel 2007, Introduction to data, Cell address, Excel Data Types, Concept of hyperlink, Introduction to formatting, number, text and date formatting, Concept of worksheet and workbook, understanding formulas, Operators in Excel 2007, Operators Precedence, Understanding Functions, Common Excel Functions such as sum, average, min, max, date, transpose, In, And, or, sqrt, power, upper, lower. Types of graphics: Word art, auto shapes, Images. Introduction to charts: overview of different types of charts available with Excel Concept of print area, margins, header, footer and other page setup options	9
IV	Using MS - PowerPoint (Latest Version): Outline of effective presentations, starting a New Presentation Files, Saving work, Creating new Slides, Working with textboxes. Changing a slides Layout, Applying a theme, Changing Colours, fonts and effects, Creating and managing custom Colour & font theme, Changing the background, Managing slides master, Managing theme, Changing the font, font size, font color, text fill, Adjusting character spacing and line spacing Formatting text boxes, Word arts, styles, Formatting bulleted lists and numbered list, Finding and replacing text, Correcting your spelling, Creating a new and editing a table's structure, Selecting, deleting, moving, copying, resizing and arranging objects. Working with drawing tools, applying shape or picture styles, Applying object borders, object fill, object effects. Working with clip art collection and modifying clip art, embed a video, Link to a video, size a video, Video playback options, Configuring a sound playback, Assigning sound to an object, Adding a digital music soundtrack, Transition effects and timings, Creating hyperlinks, Using action buttons	9
V	Basic structure of HTML:	9

	Structure of HTML Page, inserting formatting tags for Text: bold, italic, underline, line break, special character, predefine headings, paragraph, comments, Font color, size, Alignment, Margin with body tag, background and text color, Ordered and unordered lists, Use of Frames for structured viewing. Tables, Images and Links in HTML: Tables – basic structure, Using TD, TR, TH tags, use of basic elements in table: border, cellpadding, cell spacing, width, caption, align, bgcolor. Images in web page: inserting and formatting of images using SRC, border, Vspace, Hspace, align, ALT, height, width and background in HTML page. Types of links: Linking two or more web pages, linking within a web page, linking to an external page, linking to a specific point in another web page, linking image file, mailto. Working with Multimedia Objects Video and sound file. Add marquees of scrolling text. Inserting and controlling video and audio in an HTML page.	
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Text Books (TB):

TB-1	Computer Application by Er. Soumya Ranjan Behera, 1st Edition.
TB-2	Computer Fundamentals by P.K. Sinha and Priti Sinha, 8th Edition.
TB-3	https://www.tutorialspoint.com/computer_fundamentals/index.htm
TB-4	https://www.javatpoint.com/computer-fundamentals-tutorial

Other Reading (OR):

OR-1	Learning Web Design, 4th Edition by Jennifer Robbins Released August 2012.
OR-2	Using PowerPoint 2019: The Step-by-step Guide to Using Microsoft PowerPoint 2019: 3 (Using Microsoft Office), Import, 31 January 2020, by Kevin Wilso.

Relevant Website (RW):

RW-1	https://www.microsoft.com/en-in/microsoft-365/powerpoint
RW-2	https://html.com/
RW-3	https://html.spec.whatwg.org/
RW-4	https://www.tutorialspoint.com/html/index.htm

RW-5	https://www.w3schools.com/EXCEL/index.php
RW-6	https://www.javatpoint.com/excel-tutorial
RW-7	https://www.tutorialspoint.com/basics_of_computer_science/basics_of_computer_science_system.htm
RW-8	https://peda.net/kenya/ass/subjects2/computer-studies/form-1/the-computer-system

Video Links (VL):

VL-1	https://edu.gcfglobal.org/en/computerbasics/basic-parts-of-a-computer/1/
VL-2	https://www.youtube.com/watch?v=rwbho0CgEAE
VL-3	https://edu.gcfglobal.org/en/computerbasics/inside-a-computer/1/
VL-4	https://www.youtube.com/watch?v=lxq_46nY43g
VL-5	https://www.youtube.com/watch?v=lbc1HX8Jccw
VL-6	https://www.youtube.com/watch?v=bWPMSSsVdPk
VL-7	https://www.youtube.com/watch?v=HC13M8FGINc
VL-8	https://www.youtube.com/watch?v=KyKy94yHe38

MOOC Courses (MC):

Sr.No.	Platform	Topic	CO	Link	Duration
MC-1	LinkedIn Learning	Computer Literacy for Windows 10	CO1	https://www.linkedin.com/learning/computer-literacy-for-windows-10-2021/get-up-to-speed-with-your-windows-computer?autoplay=true&u=92695330	1hr 40min
MC-2	LinkedIn Learning	Introduction to Web Design and Development	CO1 CO3	https://www.linkedin.com/learning/introduction-to-web-design-and-development-14628245/web-design-and-development-in-the-code-and-no-code-era?autoplay=true&contextUrn=urn%3Ali%3AlyndaLearningPath%3A56e050fb3dd5598244f16b21&u=92695330	1 hour 50min
MC-3	LinkedIn Learning	HTML Essential Training	CO3 CO4	https://www.linkedin.com/learning/html-essential-training-4/what-is-html?autoplay=true&contextUrn=urn%3Ali%3AlyndaLearningPath%3A56e050fb3dd5598244f16b21&u=92695330	2 hour 36 min
MC-4	LinkedIn Learning	Computer Science	CO3 CO4	https://www.linkedin.com/learning/computer-science-principles-the-internet/welcome?autoplay=true&	2 hours 15 min

		Principles: The Internet		u=92695330	
MC-5	Various ISO Certified Govt. Registered Skill Development Minister of Education, Govt of India		CO3, CO4, CO5	https://iisdtd.in/product/certificate-in-computer-fundamental/ https://nios.ac.in/online-course-material/vocational-courses/certificate-in-computer-application.aspx https://sparc.org.in/courses/govt-approved/cca-certificate-in-computer-application/	
Total MOOC Course integration with Certification					12 hours

- An ISO Certified Govt. Registered Skill Development Training and Vocational Education Autonomous Organization:
<https://iisdtd.in/product/certificate-in-computer-fundamental/>
- National Institute of Open Schooling (An autonomous institute under Minister of Education, Govt of India) ISO certified:
<https://nios.ac.in/online-course-material/vocational-courses/certificate-in-computer-application.aspx>
- SARDAR PATEL Academy & Research Center-CCA (Certificate in Computer Application)
<https://sparc.org.in/courses/govt-approved/cca-certificate-in-computer-application/>

Session-wise Planning:

Module	Session	Topic	TB/RB/OR/RW/ VL	Pedagogy/ Activity planned	CO	Mode of delivery
Bride Course	Zero Lecture	Applications of Computer	VL-1 to VL-5	Blended Learning Approach		Asynchronous-Video
I	1	Basics of Computer System: Concept of Hardware and Software	TB-1 RE1 MC-1 VL-1	Open Discussion Session MCQ		Synchronous-PPT Asynchronous-Video

	2	Computer block diagram, Input Output unit		Blended Learning	CO 1	Interactive Lecture Synchronous-Video
	3	CPU, Control Unit, Arithmetic logic Unit (ALU)		PPT with MCQ		Synchronous-PPT Asynchronous-Video
	4	Memory Unit, Monitor.		Blended Learning		Synchronous-PPT Asynchronous-Video
	5	System software and Application Software, Operating system concepts		Blended Learning		Synchronous-PPT Asynchronous-Video
	6	purpose and functions, Operations of Windows OS		Blended Learning		Synchronous-PPT Asynchronous-Video
	7	Creating and naming of file and folders, copying file, renaming and deleting of files and folders		Blended Learning		Open Discussion/Debate
	8	Searching files and folders, installation application		Interactive Discussion with Demo		Synchronous-Video & Assessment
	9	creating shortcut of application on the desktop Overview of control Panel, Taskbar.		PPT Presentation & Classroom interaction		Synchronous-Video
II	10	Using MS - Word (Latest Version):	TB-1 RE1 MC-2 VL-2	Flipped Class with MCQ at the start of the class		Synchronous-Interactive Video Lecture with Quiz

		Overview of Word processor, Basics of Font type, size, color			CO 2	
	11	Effects like Bold, italic, underline, Subscript and superscript, Case changing options		Flipped Class with MCQ at the start of the class		Synchronous-Interactive Video Lecture with Quiz
	12	Inserting, deleting, undo and redo, Copy and Moving (cutting) text within a document		PPT Presentation & Classroom interaction		Synchronous - PPT
	13	Formatting Paragraphs and Lists		Blended Learning		Synchronous - PPT
	14	Setting line spacing; single, Page settings and margins including header and footer		PPT Presentation & Classroom interaction		Synchronous - PPT
	15	Setting line spacing; single, Page settings and margins including header and footer		Blended Mode with Interactive Assessment		Synchronous-Interactive Video Lecture with Quiz
	16	Inserting rows or columns, merging and splitting cells, Arithmetic Calculations in a Table		Interactive & Assessment with Report		Synchronous-Discussion, Suggestions, Assessment
	17	Working with pictures, Inserting Pictures from Files, Using		Animation Demonstration, Explanation		Synchronous-PPT Asynchronous-Video

		Drawings and WordArt				
	18	Lines and Shapes, Modifying Drawn Objects		PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video
	19	Formatting Drawn Objects, options for Creating and Modifying a WordArt Object		PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video
III	20	Using MS - Excel (Latest Version): Introduction to Excel. Introduction to data, Cell address, Excel data types	TB-1 RE1 MC-3 VL-3	PPT Presentation & Demonstration	CO 3	Synchronous-PPT Asynchronous-Video
	21	Concept of hyperlink, introduction to formatting, number, text and date formatting		PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video
	22	Concept of worksheet and workbook		PPT Presentation & Demonstration		Synchronous-PPT Asynchronous-Video
	23	Understanding formulas		PPT Presentation & Classroom interaction		Synchronous-PPT Lecture with Poll Survey
	24	Operators, Operators Precedence		PPT Presentation & Demonstration		Synchronous-PPT Asynchronous-Video
	25	Functions and Common Excel Functions, such as sum, average, min, max, date,		PPT Presentation & Classroom interaction.		Synchronous-PPT Asynchronous-Video

		transpose, In, And, or, sqrt, power, upper, lower.				
	26	Types of Graphics-Word art, auto shapes , Images		PPT Presentation & Demonstration		Synchronous-PPT Asynchronous- Quiz
	27	Introduction to charts: overview of different types of charts available with Excel		PPT Presentation & Classroom interaction		Synchronous-PPT & Interactive Video with Quiz
	28	Concept of print area, margins, header, footer and other page setup options		PPT Presentation & Classroom interaction		Synchronous-Video & Assessment
IV	29	Using MS - PowerPoint (Latest Version): outline of effective presentations, Starting a New Presentation Files, Saving work, Creating new Slides	TB-1 TB-2 RB-1 RB-2 RW-2 RW-5 VL-6 VL-7	PPT Presentation & Classroom interaction	CO 4	Synchronous-PPT Lecture with interaction
	30	Working with textboxes. Changing a slides Layout, Applying a theme, Changing Colours, fonts and effects		PPT Presentation & Demonstration		Synchronous-PPT Lecture with demo
	31	Creating and managing custom Colour & font theme, Changing the background, Managing slides master,		PPT Presentation & Classroom Interaction		Synchronous-PPT Asynchronous-Video

		Managing theme, Changing the font, font size, font color, text fill				
	32	Adjusting character spacing and line spacing Formatting text boxes, Word arts, styles, Formatting bulleted lists and numbered list, Finding and replacing text, Correcting your spelling		PPT Presentation & Problems		Synchronous-PPT Asynchronous- Video
	33	Creating a new and editing a table's structure, Selecting, deleting, moving, copying, resizing and arranging objects.		PPT Presentation & Classroom Interaction Flip Class Activity: Embed a video on PPT		Synchronous-PPT & Asynchronous- Video
	34	Working with drawing tools, Applying shape or picture styles, Applying object borders, object fill, object effects.				
	35	Working with clip art collection and modifying clip art, Embed a video, Link to a video, Size a video, Video playback options, Configuring a sound playback, Assigning sound				

		to an object, Adding a digital music soundtrack				
	36	Transition effects and timings, Creating hyperlinks, Using action buttons				
V	37-38	Basic Structure of HTML: Structure of HTML, Inserting formatting tags for text : bold, italic, underline, line break, special character, predefine headings, paragraph, comments	TB-1 TB-2 RB-1 RB-2 RW-2 RW-5 VL-8 VL-9 VL-10	PPT Presentation & Classroom interaction	CO 5	Synchronous-PPT Asynchronous- Video
	39-40	Font color, size, Alignment, Margin with body tag, background and text color, Ordered and unordered lists, Use of Frames for structured viewing.		PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video
	41-42	Tables, Images and Links in HTML: Tables – basic structure, Using TD, TR, TH tags, use of basic elements in table : border, cellpadding, cellspacing, width, caption, align, bgcolor		PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video
	43-44	Images in web		PPT		Synchronous-PPT

		page: inserting and formatting of images and background in HTML page. Types of links: Linking two or more web pages, linking within a web page, linking to an external page, linking to a specific point in another web page, linking image file, mailto.		Presentation, Demonstration & Classroom interaction		Asynchronous-Video
	45	Working with Multimedia Objects: Add marquees of scrolling text. Inserting and controlling video and audio in HTML page.		PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video Report: Creating Static Web Page

Assessment Scheme: CA: UE – 50:50

Sl. No.	Assessment Instrument	Formative/ Summative	Frequency/Stages	Weightage (%)	CO
1.	Skill Enhancement Activity-1 -Poster design and presentation	Formative	1	15	CO1, CO2,CO3
2.	Content Improvement Activity-2 -Mini Project Creating a Static Web Page		1/ 3	15	CO2,CO3, CO4, CO5
3.	Class Participation	Formative	-	5	-
4.	Periodic Test	Formative	3	15	CO1, CO2, CO3, CO4, CO5
5.	End Semester	Summative	1	50	CO1, CO2,

	Examination				CO3, CO4, CO5
Total				100	

Assessment Sheet with Rubrics for Grading & Evaluation

Activity-1 Poster design and Presentation

Sr. No.	USN No.	Student Name	On-time Submissio n	Poster design	Presentation	Total	Conversion
			5 Marks	5 Marks	5 Marks	15 Marks	15 Marks

Activity-2 Mini Project on Creating Static Web Page

Sr. No.	USN No.	Student Name	On-time Submission	Designing & Implementation of webpages	Creativity	Report with screenshot of web pages	Total	Conversion
			5 Marks	15 Marks	5 Marks	5 Marks	30 Marks	15 Marks